

|                    |                       |                       |
|--------------------|-----------------------|-----------------------|
| <b>Quiz 5B</b>     |                       |                       |
| <b>Name:</b> _____ | <b>Roll No:</b> _____ | <b>Section:</b> _____ |

|                    |                       |                       |
|--------------------|-----------------------|-----------------------|
| <b>Quiz 5B</b>     |                       |                       |
| <b>Name:</b> _____ | <b>Roll No:</b> _____ | <b>Section:</b> _____ |

**Question 1: marks are dependent on correct answer of Question 2 (10 + 5 = 15 marks)**

For the following code show the status of the stack and reflect changes in register during the execution of the program. Starting address of stack is **0300H** and starting address of memory is **0600H**

|    |                              |  |
|----|------------------------------|--|
| 1  | .data                        |  |
| 2  | array <b>byte</b> 7,8,9      |  |
| 3  | res <b>word</b> ?            |  |
| 4  | .code                        |  |
| 5  | main Proc                    |  |
| 6  | mov esi, <b>OFFSET</b> array |  |
| 7  | push esi                     |  |
| 8  | mov esi, <b>sizeof</b> array |  |
| 9  | push esi                     |  |
| 10 | mov eax,0                    |  |
| 11 | call <b>sampleproc</b>       |  |
| 12 | mov res,ax                   |  |
| 13 | mov bx,0                     |  |
| 14 | INVOKE ExitProcess,0         |  |
| 15 | main ENDP                    |  |
| 16 |                              |  |
| 17 | sampleproc proc              |  |
| 18 | Push ebp                     |  |
| 19 | mov ebp,esp                  |  |
| 20 | mov esi, [ <b>ebp+12</b> ]   |  |
| 21 | mov ecx, [ <b>ebp+8</b> ]    |  |
| 22 | cmp ecx,0                    |  |
| 23 | je L1                        |  |
| 24 | add al, [ <b>esi</b> ]       |  |
| 25 | dec ecx                      |  |
| 26 | inc esi                      |  |
| 27 | push esi                     |  |
| 28 | push ecx                     |  |
| 29 | call sampleproc              |  |
| 30 | L1:                          |  |
| 31 | pop ebp                      |  |
| 32 | ret 8                        |  |
| 33 | <b>sampleproc endp</b>       |  |

|             |           |
|-------------|-----------|
| Question 2: | (5 marks) |
|-------------|-----------|

Understand the above code (in question 1) and describe its operation in single line.

**Question 3:** (10+5+5 =20 marks)

Fill stack for your understanding and update register after every instruction. Stack address starts **0200H**

|                |                |                |
|----------------|----------------|----------------|
| <b>Quiz 5B</b> |                |                |
| Name: _____    | Roll No: _____ | Section: _____ |

|                |                |                |
|----------------|----------------|----------------|
| <b>Quiz 5B</b> |                |                |
| Name: _____    | Roll No: _____ | Section: _____ |

|                |                |                |
|----------------|----------------|----------------|
| <b>Quiz 5B</b> |                |                |
| Name: _____    | Roll No: _____ | Section: _____ |

|                |                |                |
|----------------|----------------|----------------|
| <b>Quiz 5B</b> |                |                |
| Name: _____    | Roll No: _____ | Section: _____ |

|                |                |                |
|----------------|----------------|----------------|
| <b>Quiz 5B</b> |                |                |
| Name: _____    | Roll No: _____ | Section: _____ |

|                |                |                |
|----------------|----------------|----------------|
| <b>Quiz 5B</b> |                |                |
| Name: _____    | Roll No: _____ | Section: _____ |